

HW 2 – The Addition Rule

A standard deck of cards has 52 cards. Each deck has 4 different suits: Clubs (black), Spades (black), Diamonds (red), and Hearts (red). Each suit contains the following cards: Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, and a King.

1) Consider the events listed below and a fair deck of cards.

C = draw a heart

D = draw a face card

E = draw an ace

Which of the following are mutually exclusive? Explain.

a) C and D

b) C and E

c) D and E

2) One card is randomly selected from a standard deck of 52 cards. What is the probability the card will be ...

a) 10 *or* Face Card

b) 7 *or* Diamond

3) One six-sided die is rolled. What is the probability that the die will be ...

a) Factor of 12 *or* Factor of 9

b) Less than 3 *or* Greater than 5

4) A recent study of 200 nurses found that of 125 female nurses, 56 had bachelor's degrees; and of 75 male nurses, 34 had higher than a bachelor's degrees. Create a two-way table with that information below.

	Bachelor's Degree	Higher than a Bachelor's Degree	Total
Female			
Male			
Total			

a) What is the probability of choosing a female nurse or a nurse with a bachelor's degree?

b) What is the probability of choosing a male nurse or a nurse with a bachelor's degree?

5) The probability that a student owns a car is 0.65, and the probability that a student owns a computer is 0.82. If the probability that a student owns both is 0.55, what is the probability that a randomly selected student owns a car or computer?

6) At a school with 200 male students, 58 play football, 40 play basketball and 8 play both. Find the probability that a randomly selected male student plays neither sport.